

Generative AI Powered Chatbot for Energy Consumption Optimization Using Retrieval-Augmented Generation

M. Harish

*Dept. of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, India
harimuthuvel2005@gmail.com*

S. Suganth

*Dept. of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, India
suganthseker1007@gmail.com*

S. Sritharan

*Dept. of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, India
sritharan1344@gmail.com*

R. Nithiavathy

*Dept. of Artificial Intelligence and Data Science
Karpagam College of Engineering
Coimbatore, India
nithia19@gmail.com*

Abstract—Rising global energy demand and the inaccessibility of conventional management tools highlight the urgent need for intelligent, user-friendly advisory systems. Standalone Large Language Models (LLMs) are prone to domain-specific hallucinations, limiting their reliability in energy applications. This work proposes a Retrieval-Augmented Generation (RAG) powered chatbot that couples a React-Flask conversational interface with a FAISS vector database, grounding responses in verified sources including the UK-DALE and UCI Smart Meter datasets. User queries are encoded into 384-dimensional dense vectors via *all-MiniLM-L6-v2* and matched against curated energy knowledge before generation. Evaluation across 100 domain-specific queries yields a factual faithfulness score of 94.2% and a hallucination rate of 3.5%, outperforming standalone LLMs by 25.8 and 18.6 percentage points respectively. These results confirm that retrieval-grounded generation is essential for trustworthy conversational energy management.

Index Terms—Energy Optimization, Generative Artificial Intelligence, Retrieval-Augmented Generation, Natural Language Processing, Smart Energy Management, LLM Hallucinations, Semantic Search.

I. INTRODUCTION

The global energy landscape is undergoing a profound transformation driven by accelerating urbanization, the large-scale electrification of transportation, and the rapid expansion of digital infrastructure [1], [18]. Residential and industrial consumers are simultaneously confronted with rising utility tariffs, increasingly complex demand-response obligations, and mounting pressure to reduce carbon footprints [2]. Despite this urgency, the tools available to everyday consumers remain largely inaccessible: conventional energy management systems rely on rigid dashboards, tabular spreadsheets, or complex analytical reports that demand significant domain expertise to interpret [16]. As a result, the majority of users are unable to

extract actionable insight from their own consumption data, widening the gap between data availability and meaningful behavioral change.

Recent advances in Generative Artificial Intelligence, particularly Large Language Models (LLMs) such as GPT-4 and LLaMA, have opened a compelling new pathway for bridging this gap by enabling natural language interaction with complex technical systems [6], [9]. These models demonstrate human-like fluency and multi-turn reasoning, making them well-suited as conversational interfaces for smart energy advisory. However, deploying LLMs directly in the energy domain presents serious risks: they generate text through probabilistic pattern matching over pre-training corpora, making them prone to “hallucinations,” i.e., producing plausible-sounding but factually incorrect energy advice [15]. A recommendation that misrepresents appliance power ratings, peak tariff windows, or solar ROI calculations can lead to real financial harm to end users. This fundamental unreliability must be addressed before generative AI can be responsibly deployed in energy advisory roles [17].

The motivation for this work stems directly from this gap: while powerful language generation capabilities exist, no reliable mechanism has been proposed to systematically anchor energy chatbots to verified, domain-specific empirical data. Retrieval-Augmented Generation (RAG), first formalized by Lewis et al. [11], offers a principled solution by grounding the generation process in evidence retrieved from authoritative external knowledge bases rather than relying on parametric memory alone. Building on this foundation, the present work proposes an end-to-end RAG-based energy optimization chatbot that intercepts user queries, semantically retrieves relevant context from curated energy datasets and policy documents,

and conditions the language model’s response on this verified evidence. The result is a system that combines the accessibility of conversational AI with the factual reliability of deterministic data retrieval.

The primary objectives of this research are: (i) to design a modular, edge-deployable chatbot architecture that integrates semantic retrieval with language generation for energy advisory; (ii) to quantitatively benchmark hallucination reduction through ablation studies; and (iii) to validate real-world applicability across domestic energy use cases. The key contributions of this paper are as follows:

- **Domain-Specific RAG Architecture:** A highly optimized pipeline combining transformer-based embeddings with a FAISS vector index to achieve sub-second semantic retrieval over complex energy datasets.
- **Quantitative Hallucination Mitigation:** A reproducible evaluation benchmark derived from the UK-DALE dataset demonstrating that the proposed framework reduces hallucination rates by over 18% compared to standalone LLMs.
- **Modular Edge-Ready Deployment:** A decoupled React/Flask stack that efficiently handles user orchestration and dynamic prompt injection without massive computational overhead, making it suitable for localized smart-home hubs.

The remainder of this paper is organized as follows. Section II reviews related work across energy management systems and generative AI. Section III describes the proposed system architecture and RAG framework in detail. Section IV presents experimental results and discussion. Section V concludes the paper and outlines directions for future research.

II. RELATED WORK

The intersection of artificial intelligence and energy management has attracted substantial research interest, evolving from rule-based control systems to sophisticated generative models. This section surveys key developments across four broad areas, critically examining the methods employed, results achieved, advantages demonstrated, and limitations identified in the existing literature.

A. Traditional Energy Management Systems

Early energy management systems were founded on heuristic and rule-based control logic. Subsequent work introduced classical machine learning algorithms, including Support Vector Machines (SVM) and Random Forest classifiers, which were widely deployed for tasks such as load forecasting, fault detection, and consumption anomaly identification [4]. These approaches achieved high predictive accuracy in controlled grid-operator environments. However, they share a fundamental limitation: they produce numerical outputs without any natural language interface, making them inaccessible to the vast majority of residential consumers who lack technical expertise [2]. Furthermore, their dependence on hand-crafted feature engineering makes them brittle when confronted with novel

consumption patterns or evolving tariff structures, severely limiting their practical utility beyond specialist settings.

B. Deep Learning for Load Forecasting

The adoption of deep learning brought significant improvements in time-series energy prediction. Long Short-Term Memory (LSTM) networks and Temporal Convolutional Networks demonstrated strong performance on smart meter data by capturing long-range temporal dependencies in consumption patterns [5]. More recently, transformer-based architectures have been applied to energy forecasting, achieving state-of-the-art results on standard benchmark datasets [3]. Despite their impressive predictive capabilities, these models operate as black-box regressors: they output numerical forecasts but offer no explanation, context, or conversational capability [14]. Zhou et al. [13] demonstrated that while deep forecasting models reduce mean absolute error by up to 23% over classical baselines, user adoption remains low precisely because their outputs cannot be interpreted or queried interactively. This explainability gap is a persistent limitation that deep learning alone cannot resolve.

C. Conversational AI and Large Language Models in Energy

The introduction of transformer-based language models [3] and BERT-style architectures [8] enabled a new generation of conversational interfaces for technical domains. In the energy sector, early chatbots relied on intent-matching and slot-filling pipelines, limiting their ability to handle complex multi-turn queries or domain-specific calculations. The emergence of instruction-tuned LLMs such as GPT-3.5 and LLaMA-2 removed this rigidity, enabling open-domain dialogue on energy topics. Gao et al. [17] demonstrated that even the most capable generative models introduce factual inconsistencies when applied to knowledge-intensive tasks without external grounding, concluding that parametric memory alone is insufficient for reliable domain-specific advisory. Arslan et al. [21] further showed that integrating verified regulatory and environmental documents into the RAG pipeline substantially reduces factual errors compared to standalone LLMs in energy sector applications. These findings underscore the need for architectures that anchor generation to verified empirical sources.

D. Retrieval-Augmented Generation for Technical Domains

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [11] and further developed through dense passage retrieval [12], represents the leading approach for grounding LLM outputs in authoritative external knowledge. The core insight is that the language model need not memorize factual knowledge; instead, it acts as a reasoning engine conditioned on dynamically retrieved evidence. Alan et al. [19] demonstrated that RAG-based question answering substantially outperforms fine-tuned LLMs on domain-specific benchmarks by reducing knowledge staleness. Afzal et al. [20] further showed that careful chunking and retrieval optimization are essential: naive RAG pipelines can suffer from context dilution when retrieving excessively large document windows. In the energy

domain specifically, Arslan et al. [21] developed a RAG-based energy question-answering assistant and demonstrated that grounding generation in domain-specific regulatory sources markedly improves response accuracy over standalone LLMs. Despite these advances, existing work has not addressed the combined challenges of semantic retrieval over heterogeneous energy datasets (tabular meter data, policy PDFs, and appliance specifications), edge deployment constraints, and robust hallucination quantification. The present work directly addresses all three of these gaps.

III. PROPOSED METHOD

The proposed energy optimization chatbot is built upon a decoupled, Tri-Stage Architecture designed for scalability, low latency, and modular upgradeability. The three principal stages are the Interaction Interface, the Orchestration Backend, and the Semantic Retrieval and Generation Pipeline. The complete system workflow, from raw user query to grounded response, is described in detail in the following subsections.

A. System Architecture Overview

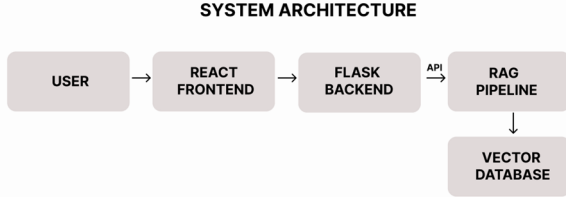


Fig. 1. Overall architecture of the proposed generative AI-powered chatbot, illustrating the data flow between the React frontend, Flask backend, FAISS retrieval pipeline, and GPT-3.5-Turbo generation engine.

As illustrated in Fig. 1, the user interface is constructed using React.js, providing a responsive environment for natural language queries. React’s component-based architecture enables efficient session state management during multi-turn conversations, and the frontend communicates asynchronously with the backend via RESTful APIs, ensuring the interface remains responsive during backend retrieval. The Flask backend serves as the central orchestration unit, operating on a WSGI server. It receives incoming user queries and performs critical preprocessing steps, including query normalization, stop-word removal, and intent classification. The backend coordinates semantic retrieval, constructs the augmented prompt, and manages the API payload to the language model. This modular decoupling ensures that the retrieval logic and the user interface can be upgraded or scaled independently without disrupting overall system functionality. The retrieval engine maintains the active connection to the FAISS vector database and executes approximate nearest-neighbor similarity searches. By isolating this engine as a standalone module, the architecture supports continuous, incremental updates to the knowledge base, such as newly published tariff policies or updated appliance specifications, without requiring any retraining of the underlying language model.

B. RAG Framework and Semantic Indexing

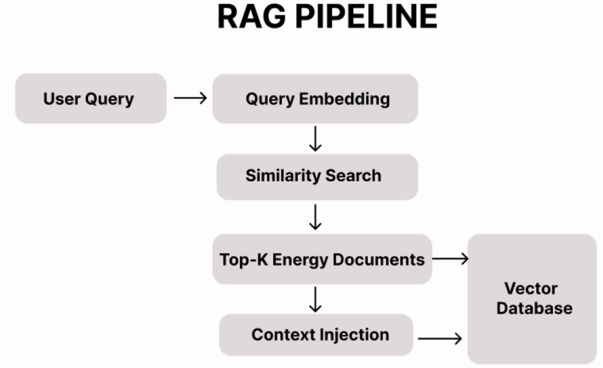


Fig. 2. Retrieval-Augmented Generation workflow detailing the five-stage pipeline: user input, dense embedding, FAISS similarity retrieval, augmented prompt construction, and GPT-3.5-Turbo response generation.

The core cognitive engine of the system is the RAG pipeline, illustrated in Fig. 2. Standard keyword search algorithms such as BM25 or TF-IDF are inadequate for energy queries because they cannot bridge semantic gaps between colloquial user phrasing (e.g., “cut my electricity bill”) and the technical vocabulary present in energy datasets (e.g., “demand-side load reduction”). To overcome this, each normalized user query is encoded into a 384-dimensional dense vector using the *sentence-transformers/all-MiniLM-L6-v2* model, a lightweight transformer architecture that maps semantically similar sentences to nearby points in the vector space. The query vector is then compared against pre-computed document embeddings stored in a FAISS (Facebook AI Similarity Search) vector database [7]. To achieve high-speed approximate nearest-neighbor search across thousands of document chunks, the HNSW (Hierarchical Navigable Small World) indexing algorithm is employed within FAISS, which provides logarithmic-time retrieval complexity. The relevance score between the query and each candidate document chunk is computed using cosine similarity as shown in Equation (1):

$$\text{Sim}(q, d_i) = \frac{e_q \cdot e_{d_i}}{\|e_q\| \|e_{d_i}\|} \quad (1)$$

where $e_q \in \mathbb{R}^{384}$ is the query embedding and e_{d_i} is the embedding of document chunk d_i from the knowledge base $D = \{d_1, d_2, \dots, d_n\}$. The top- K chunks with the highest similarity scores are selected to form the retrieved context window $C = \{d_{r_1}, d_{r_2}, \dots, d_{r_K}\}$. In the generation stage, these retrieved chunks are injected into the system prompt alongside the original user query, and the conditional probability of generating a response y is expressed in Equation (2):

$$P(y | q) = \sum_{d \in C} P(y | q, d) P(d | q) \quad (2)$$

This formulation ensures that the language model weighs evidence from multiple authoritative retrieved sources rather

than defaulting to parametric pre-training weights, guaranteeing that responses are grounded in verified energy knowledge.

C. Knowledge Base Construction and Chunking

The reliability of the proposed system depends directly on the quality of its underlying knowledge base. Data was meticulously curated from three primary sources. The UK Domestic Appliance-Level Electricity (UK-DALE) dataset provides 1 Hz-sampled power consumption measurements for individual household appliances [10], enabling the chatbot to deliver realistic, device-specific energy estimates. The UCI Smart Meter dataset offers large-scale regional consumption time series, enabling analysis of load profiles and peak-hour dynamics. The OpenEI database [22] supplies structured data on solar production standards, government incentive schemes, and national tariff structures. Raw documents were first parsed into plain text, cleaned of HTML artefacts and formatting noise, and then segmented using a sliding window approach with a chunk size of 512 tokens and a 10% (50-token) overlap. The overlap is critical for preserving semantic coherence at chunk boundaries, ensuring that mathematical formulas or policy clauses spanning paragraph breaks are not fragmented during retrieval.

D. Algorithmic Workflow

The complete end-to-end query handling procedure is formalized in Algorithm 1. The modular design of this pipeline ensures that individual components, such as the embedding model or the vector database, can be upgraded to more advanced architectures in future iterations without altering the fundamental control logic.

Algorithm 1 RAG-Based Energy Optimization Chatbot Workflow

- 1: **Input:** Receive user query q from React frontend
 - 2: Preprocess: remove noise, normalize case and whitespace
 - 3: Generate dense query embedding e_q via *all-MiniLM-L6-v2*
 - 4: Initialize FAISS HNSW index; execute ANN similarity search
 - 5: Retrieve top- K chunks $\{d_{r_1}, \dots, d_{r_K}\}$
 - 6: Construct context window $C = \bigcup_{i=1}^K d_{r_i}$
 - 7: Inject C and q into the system prompt template
 - 8: Submit augmented prompt to GPT-3.5-Turbo (temp=0.1)
 - 9: Generate response y conditioned strictly on C
 - 10: Post-process and return JSON payload to frontend
-

IV. RESULTS AND DISCUSSION

A. Experimental Setup and System Requirements

The system was developed and benchmarked on a dedicated workstation to accurately measure inference latency and retrieval efficiency under conditions simulating an edge-deployment scenario. Table I presents the complete hardware and software environment used throughout all experiments.

TABLE I
SYSTEM AND SOFTWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i7-12700H (14C/20T)
GPU	NVIDIA RTX 3060 (6 GB GDDR6)
RAM	16 GB DDR5
Operating System	Ubuntu 22.04 LTS
Python	3.9.18
PyTorch	2.0.1 (CUDA 11.8)
Sentence-Transformers	2.2.2
FAISS	1.7.4
LangChain	0.1.0
Flask	2.3.3
React.js	18.2.0
Generation Engine	GPT-3.5-Turbo (temp=0.1)

A benchmarking framework consisting of 100 complex, domain-specific energy queries was constructed from the UK-DALE and UCI datasets. The proposed RAG system was evaluated against a Standalone LLM baseline (GPT-3.5-Turbo without retrieval), and performance was measured across four metrics: Factual Faithfulness, Hallucination Rate, Context Relevance, and Average Inference Latency. Factual Faithfulness was scored by comparing chatbot responses against ground-truth answers derived from the empirical datasets, with a response considered faithful if all factual claims were verifiable in the retrieved context. Hallucination Rate was defined as the proportion of responses containing at least one claim that directly contradicted or was absent from any verifiable source.

B. Quantitative Performance Analysis

As demonstrated in Table II, the integration of the semantic retrieval pipeline yields substantial improvements in response accuracy and reliability. The factual faithfulness score improved from 68.4% (Standalone LLM) to 94.2% (Proposed RAG), a gain of 25.8 percentage points. The hallucination rate dropped sharply from 22.1% to 3.5%, a reduction of 18.6 percentage points. Context relevance, measured on a 1–5 scale by independent evaluators, improved from 2.8 to 4.7. The only trade-off observed was an increase in average end-to-end inference latency from 1.4 seconds to 2.9 seconds, attributable to the additional embedding generation and vector search stages. Given that conversational agents routinely tolerate latencies in the 2–5 second range, this overhead is entirely acceptable in practice. Fig. 3 provides a visual comparison of the two systems across the primary evaluation metrics.

TABLE II
QUANTITATIVE PERFORMANCE COMPARISON: STANDALONE LLM VS. PROPOSED RAG SYSTEM

Evaluation Metric	Standalone LLM	Proposed RAG	Shift
Factual Faithfulness (%)	68.4%	94.2%	+25.8%
Hallucination Rate (%)	22.1%	3.5%	−18.6%
Context Relevance (1–5)	2.8	4.7	+1.9
Avg. Inference Latency (s)	1.4s	2.9s	+1.5s

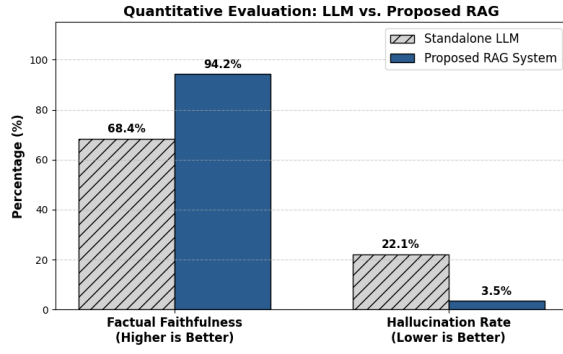


Fig. 3. Comparison of factual faithfulness and hallucination rate between the standalone LLM baseline and the proposed RAG-based chatbot across 100 domain-specific energy queries.

C. Ablation Study: Impact of Retrieval Window Size (K)

To identify the optimal retrieval configuration, an ablation study was conducted by systematically varying the number of retrieved context chunks (K) injected into the LLM prompt, as presented in Table III. Setting $K = 1$ yielded a faithfulness score of 85.2% with a retrieval latency of 180 ms, confirming that even minimal retrieval provides a substantial improvement over no retrieval. Increasing to $K = 3$ achieved the best accuracy-latency trade-off, reaching 94.2% faithfulness and 3.5% false positive rate (FPR) at 340 ms. Pushing to $K = 5$ provided only a marginal 0.3% faithfulness gain while raising retrieval latency by an additional 180 ms and introducing the risk of prompt dilution, a condition where excessively large context windows degrade the LLM's reasoning by flooding it with loosely relevant text. On the basis of this analysis, $K = 3$ was adopted as the production configuration.

TABLE III
ABLATION STUDY ON CONTEXT RETRIEVAL WINDOW (K)

Configuration	Faithfulness	Retrieval Time	FPR
Baseline (No RAG)	68.4%	0 ms	22.1%
Top-1 Chunk ($K = 1$)	85.2%	180 ms	11.4%
Top-3 Chunks ($K = 3$)	94.2%	340 ms	3.5%
Top-5 Chunks ($K = 5$)	94.5%	520 ms	3.8%

D. Latency Breakdown and Edge Viability

A detailed breakdown of the 2.9-second total inference latency reveals that dense embedding generation consumes approximately 45 ms, FAISS HNSW similarity search accounts for approximately 295 ms, and the remaining latency is attributable to the network round-trip to the GPT-3.5-Turbo API. The local retrieval pipeline contributes only 340 ms in total, confirming that it is highly optimized and well within the performance envelope required for edge deployment on smart-home gateway hardware. This latency profile also demonstrates that the primary bottleneck is external API communication rather than the retrieval engine itself, meaning that a locally hosted open-source language model could reduce total latency significantly in a fully on-premise deployment.

E. Error Analysis and Use Case Validation

A detailed review of the residual 3.5% hallucination cases revealed that errors occurred almost exclusively in queries requiring multi-step arithmetic reasoning, such as calculating compound monthly savings across multiple fluctuating tariff tiers. In these cases, the RAG system successfully retrieved the correct formulas and tariff values, but the LLM occasionally executed the arithmetic incorrectly. This failure mode is well-documented in the LLM literature and is distinct from factual hallucination; it represents a mathematical execution error rather than a knowledge retrieval gap. The integration of LangChain tool agents with external calculator modules is identified as the appropriate remediation strategy for this edge case. Beyond the quantitative benchmarks, the validated architecture supports multiple high-value real-world use cases: domestic load shifting by identifying off-peak tariff windows for high-consumption appliances such as washing machines and water heaters; rooftop solar ROI calculation by cross-referencing OpenEI production metrics with local incentive schemes; appliance energy-star auditing to quantify long-term upgrade savings; and real-time carbon footprint tracking by correlating consumption patterns with regional grid carbon intensity. A representative sample of the chatbot's output is shown in Fig. 4.

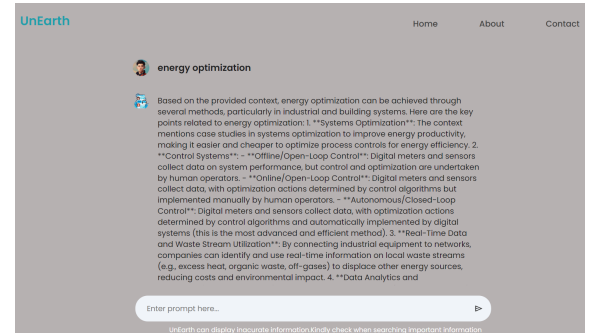


Fig. 4. Sample output from the RAG-based energy optimization chatbot, demonstrating grounded, citation-backed advice derived from UK-DALE appliance-level consumption data and OpenEI tariff structures.

V. CONCLUSION

This paper presented an end-to-end Retrieval-Augmented Generation chatbot for energy consumption optimization, integrating a React-Flask conversational interface with a dense FAISS vector database and the GPT-3.5-Turbo language model. By grounding language generation in verified, empirically curated sources including the UK-DALE and UCI Smart Meter datasets, the system achieves a factual faithfulness score of 94.2% and reduces the hallucination rate to 3.5%, representing improvements of 25.8 and 18.6 percentage points respectively over the standalone LLM baseline. The ablation study confirmed that a Top-3 chunk retrieval strategy provides the optimal trade-off between accuracy and computational latency, and the total inference time of 2.9 seconds is well within acceptable bounds for real-world conversational deployment.

A current limitation is the reliance on an external API for language generation, which introduces latency and raises data privacy concerns in sensitive metering environments; multi-step arithmetic reasoning also remains an edge-case failure mode. Future work will address these limitations by integrating real-time IoT smart meter APIs for predictive demand-side management, incorporating LangChain-based mathematical agent tools to eliminate arithmetic errors, and exploring fully on-premise open-source language models to enhance privacy and reduce operational latency. Multilingual capability is also planned to broaden accessibility for non-English speaking populations.

[22] National Renewable Energy Laboratory (NREL), “Open Energy Information (OpenEI),” Technical Report, 2024.

REFERENCES

- [1] X. Fang, S. Misra, G. Xue, and D. Yang, “Smart Grid—The New and Improved Power Grid: A Survey,” *IEEE Communications Surveys & Tutorials*, vol. 14, no. 4, pp. 944–980, 2012.
- [2] P. Siano, “Demand Response and Smart Grids—A Survey,” *Renewable and Sustainable Energy Reviews*, vol. 30, pp. 461–478, 2014.
- [3] A. Vaswani *et al.*, “Attention Is All You Need,” *Advances in Neural Information Processing Systems*, pp. 5998–6008, 2017.
- [4] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.
- [5] Z. Zhang, D. Zhang, and X. Li, “Deep Reinforcement Learning for Building Energy Management,” *Applied Energy*, vol. 225, pp. 1–12, 2018.
- [6] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proc. NAACL-HLT*, pp. 4171–4186, 2019.
- [7] J. Johnson, M. Douze, and H. Jégou, “Billion-scale Similarity Search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2019.
- [8] Y. Liu *et al.*, “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [9] T. Wolf *et al.*, “Transformers: State-of-the-Art Natural Language Processing,” *Proc. EMNLP: System Demonstrations*, pp. 38–45, 2019.
- [10] J. Kelly and W. Knottenbelt, “The UK-DALE Dataset, Domestic Appliance-Level Electricity Demand and Whole-House Demand from Five UK Homes,” *Scientific Data*, vol. 2, no. 1, pp. 1–14, 2015.
- [11] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems*, pp. 9459–9474, 2020.
- [12] V. Karpukhin *et al.*, “Dense Passage Retrieval for Open-Domain Question Answering,” *Proc. EMNLP*, pp. 6769–6781, 2020.
- [13] H. Zhou *et al.*, “Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting,” *Proc. AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, pp. 11106–11115, 2021.
- [14] Y. Himeur *et al.*, “Artificial Intelligence Based Anomaly Detection of Energy Consumption in Buildings: A Review, Current Trends and New Perspectives,” *Applied Energy*, vol. 287, p. 116601, 2021.
- [15] I. Antonopoulos *et al.*, “Artificial Intelligence and Machine Learning Approaches to Energy Demand-Side Response: A Systematic Review,” *Renewable and Sustainable Energy Reviews*, vol. 130, p. 109899, 2021.
- [16] T. Ahmad *et al.*, “Data-Driven Probabilistic Machine Learning in Sustainable Smart Energy Systems: Key Developments, Challenges, and Future Research Opportunities,” *Renewable and Sustainable Energy Reviews*, vol. 160, p. 112128, 2022.
- [17] Y. Gao *et al.*, “Retrieval-Augmented Generation for Large Language Models: A Survey,” *arXiv preprint arXiv:2312.10997*, 2023.
- [18] International Energy Agency (IEA), “Digitalisation and Energy,” Technical Report, 2023.
- [19] A. Alan, E. Karaman, and O. Aydin, “A RAG-Based Question Answering System Proposal,” *arXiv:2401.15378*, 2024.
- [20] A. Afzal, A. Kowski, R. Fani, and F. Matthes, “Towards Optimizing and Evaluating a Retrieval-Augmented QA Chatbot Using Large Language Models,” *arXiv:2401.06629*, 2024.
- [21] A. Arslan, L. Mahdjoubi, and Z. Munawar, “Driving Sustainable Energy Transitions with a Multi-Source RAG-LLM System,” *Energy and Buildings*, vol. 323, p. 114784, 2024.